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ECS.101 - Assignment 4

1. $x(t) = u(t) + e^{-2(t-1)}u(t-1) + e^{-2(t+1)}u(t+1)$

We know, $e^{-at}u(t) \xleftrightarrow{LT} \frac{1}{s+a}$; $Re(s) > -a$
$[a \in \mathbb{R}]$ — (1)

$\therefore x(t) \xleftrightarrow{LT} X(s)$

$\Rightarrow X(s) = \frac{1}{s} + \frac{e^{-s}}{s+2} + \frac{e^{s}}{s+2}$;

$Re(s) > 0$ & $Re(s) > -2$ & $Re(s) > -2$

[By linearity of LT and (1)]

$\therefore x(t) \xleftrightarrow{LT} X(s) = \frac{1}{s} + \frac{(e^{s}+e^{-s})}{s+2}$

; ROC : $Re(s) > 0$.

$[Re(s) < 0]$

Now, given $X(s)$, let ROC be : $Re(s) < -2$.

Also, $-e^{-at}u(-t) \xleftrightarrow{LT} \frac{1}{s+a}$; $Re(s) < -a$ — (2)

$\cancel{\Rightarrow x_1(t) \xleftrightarrow{LT} X_1(s) = \frac{1}{s} + \frac{e^{-s}}{s+2} + \frac{e^{s}}{s+2}}$

$\Rightarrow x_1(t) \xleftrightarrow{LT} X_1(s) = \frac{1}{s} + \frac{e^{-s}}{s+2} + \frac{e^{s}}{s+2}$

; $x_1(t) = \boxed{-u(-t) - e^{-2(t+1)}u(-(t+1)) - e^{-2(t-1)}u(-(t-1))}$

[From (2)]

[Also, $x(t) \xleftrightarrow{LT} X(s) \Rightarrow x(t-t_0) \xleftrightarrow{LT} X(s)e^{-st_0}$

Again for same $X(s)$, let ROC be : $-2 < Re(s) < 0$

$\Rightarrow x_2(t) \xleftrightarrow{LT} \frac{1}{s} + \frac{e^{-s}}{s+2} + \frac{e^{s}}{s+2}$

$\therefore x_2(t) = \boxed{-u(-t) + e^{-2(t-1)}u(t-1) + e^{-2(t+1)}u(t+1)}$

[From 1 & 2]

Because there were three possible ROCs (from 2 poles), given the function $X(s)$, one can get 3 f^ns in time-domain.

2. a) Poles at $s = -1, -2$, where $D(s) = 0$.
Zeroes at $s = \pm 2j$, where $N(s) = 0$.

$\therefore X(s) = \frac{(s-2j)(s+2j)}{(s+1)(s+2)} = \frac{N(s)}{D(s)}$

$\Rightarrow X(s) = \frac{s^2+4}{(s+1)(s+2)}$

b) 3 distinct signals in the time-domain can have the given laplace transform. Their ROCs are as:

→ $Re(s) > -1$

→ $-2 < Re(s) < -1$

→ $Re(s) < -2$

3. $x(t) = \sum_{k=0}^{\infty} \delta(t-kt_0), \quad t_0 > 0$

$\Rightarrow \cancel{x(t) = \delta(t}$

$\Rightarrow x(t) \xleftrightarrow{LT} \boxed{X(s) = \sum_{k=0}^{\infty} e^{-kst_0}}$

$[\because \delta(t) \xleftrightarrow{LT} 1$ and $x(t-\lambda) \xleftrightarrow{LT} e^{-s\lambda} X(s)]$

; ROC : complete s-plane, i.e, $\boxed{Re(s) \in \mathbb{R}}$

Due to linearity of LT, the Σ is conserved.
Also, since $X(s)$ is a GP series (tending to infinity)

$$X(s) = \sum_{k=0}^{\infty} e^{-kst_0} = \frac{e^{0}}{1-e^{-st_0}} = \frac{1}{1-e^{-st_0}}, \; t_0 > 0.$$

4. a) $g(t) = x(t) + \alpha x(-t)$
$x(t) = \beta e^{-t} u(t)$

$g(t) \xleftrightarrow{LT} G(s) = \frac{s}{s^2-1}, \quad -1 < Re(s) < 1$

$\Rightarrow g(t) = \beta e^{-t} u(t) + \alpha\beta e^{t} u(-t)$

$\Rightarrow G(s) = \frac{\beta}{s+1} + \alpha\beta \int_{-\infty}^{\infty} e^{-st} . e^{t} u(-t)\, dt, \; Re(s) > -1$

$(\because e^{-at} u(t) \xleftrightarrow{LT} \frac{1}{s+a}, \; Re(s) > -a)$

$= \frac{\beta}{s+1} + \alpha\beta \int_{-\infty}^{0} e^{t(-s+1)} dt \quad [\because u(-t) = 0 \; \forall t > 0]$

$= \frac{\beta}{s+1} + \frac{\alpha\beta e^{t(-s+1)}}{-s+1} \Big|_{t=-\infty}^{t=0} = \frac{\beta}{s+1} + \frac{\alpha\beta}{-s+1}$

; $Re(s) > -1$ & $Re(s) < 1$

$\Rightarrow -1 < Re(s) < 1$

$\therefore$ By comparison, $\dfrac{\beta}{s+1} + \dfrac{\alpha\beta}{1-s} = \dfrac{s}{s^2-1}$

$\cancel{\Rightarrow \dfrac{\beta(1+\alpha)}{1-s^2} = \dfrac{s}{s^2-1}}$

$\cancel{\Rightarrow \beta(1+\alpha) = -s}$

$\Rightarrow \dfrac{\beta(1-s) + \alpha\beta(1+s)}{1-s^2} = \dfrac{-s}{1-s^2}$

$\Rightarrow s(\alpha\beta - \beta) + (\alpha\beta + \beta) = -s$

Comparing coefficients,

$\alpha\beta - \beta = -1$; $\alpha\beta + \beta = 0 = \beta(\alpha+1)$

$\Rightarrow \beta(1-\alpha) = 1$; $\beta = 0$ or $\alpha = -1$

$\beta = 0 \Rightarrow 0(1-\alpha) = 0 \neq 1 \Rightarrow \beta \neq 0$

$\therefore \boxed{\alpha = -1} \Rightarrow \beta(1+1) = 2\beta = 1 \Rightarrow \boxed{\beta = 1/2}$

b). $x(t) \xleftrightarrow{LT} X(s)$

$x(t) = \dfrac{1}{2\pi j} \displaystyle\int_{\sigma - j\omega}^{\sigma + j\omega} X(s)\, e^{st}\, ds$

$\Rightarrow x^*(t) = \dfrac{1}{2\pi(-j)} \displaystyle\int_{\sigma + j\omega}^{\sigma - j\omega} X^*(s)\, e^{s^* t}\, ds^*$

$= \dfrac{1}{2\pi j} \displaystyle\int_{\sigma - j\omega}^{\sigma + j\omega} X^*(s)\, e^{s^* t}\, ds^*$

$\cancel{\text{Let } s = x + iy \Rightarrow s^* = x - iy \; (x, y \in \mathbb{R})}$

$\cancel{\Rightarrow ds = dx + i\,dy \; ; \; ds^* = dx - i\,dy}$

$\cancel{\text{But, } s \in (\sigma - j\omega, \sigma + j\omega) \Rightarrow Re(s) = \sigma}$

$\cancel{\Rightarrow ds = i\,dy \; ; \; ds^* = -i\,dy}$

~~But since dy is a differential variable, ds = ds~~

$\Rightarrow x^*(t) = \dfrac{1}{2\pi j} \displaystyle\int_{\sigma - j\omega}^{\sigma + j\omega} X^*(s)\, e^{s^* t}\, ds^*$

$[z = s^*]$

$$\Rightarrow x^*(t) = \frac{1}{2\pi j}\int_{\sigma-j\omega}^{\sigma+j\omega} X^*(z^*)\, e^{zt}\, \underbrace{dz}_{\text{dummy var.}}$$

But, $x(t)$ is real-valued $\Rightarrow x(t) = x^*(t)$.

$$\Rightarrow \frac{1}{2\pi j}\int_{\sigma-j\omega}^{\sigma+j\omega} X(s)\, e^{st}\, ds = \frac{1}{2\pi j}\int_{\sigma-j\omega}^{\sigma+j\omega} X^*(z^*)\, e^{zt}\, dz$$

$$\Rightarrow \boxed{X(s) = X^*(s^*)}$$

- If $X(s)$ has a pole at $s = s_0$,

$$\left.\frac{1}{X(s)}\right|_{s=s_0} = 0.$$

Also, $X(s) = X^*(s^*)$

$$\Rightarrow X(s_0) = X^*(s_0^*)$$

$$\Rightarrow \frac{1}{X^*(s_0^*)} = \frac{1}{X(s_0)} = 0$$

$$\Rightarrow \left(\frac{1}{X^*(s_0^*)}\right)^* = 0^* \Rightarrow \frac{1}{X(s_0^*)} = 0$$

$\Rightarrow X(s)$ has a pole at $s = s_0^*$.

Similarly, if $X(s)$ has a zero at $s = s_0$,

$X(s_0) = 0 \Rightarrow X^*(s_0^*) = 0$

$\Rightarrow (X^*(s_0^*))^* = 0^*$

$\Rightarrow X(s_0^*) = 0$

$\Rightarrow X(s)$ has a zero at $s = s_0^*$.

c). $\phi_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)\,x(t+\tau)\,dt$

$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$ —(1)

And, $\phi_{xx}(t) = \int_{-\infty}^{\infty} x(\tau)\,\cancel{x}(t+\tau)\,\cancel{dt}\,d\tau$ —(2)

[dummy var.]

(1) & (2) are equal

$\Rightarrow \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau = \int_{-\infty}^{\infty} x(\tau)\,\cancel{x}(t+\tau)\,d\cancel{t}$

~~$\Rightarrow h(t-\tau) = x(t+\tau)$~~

~~$\Rightarrow h(t) = x$ Put t~~

$\Rightarrow h(t-\tau) = x(t+\tau)$

Put $t=0 \Rightarrow h(-\tau) = x(\tau)$

$\therefore \boxed{h(t) = x(-t)}$

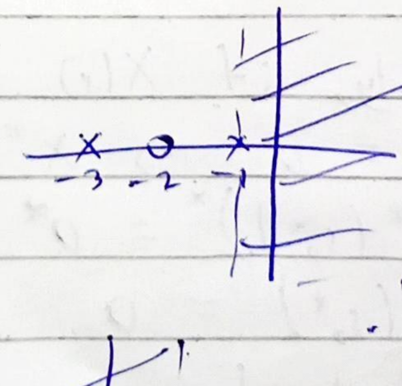
• $\phi_{xx}(t) = \cancel{h(t)}\; x(t) * h(t) = x(t) * x(-t)$

Applying Laplace Transform:

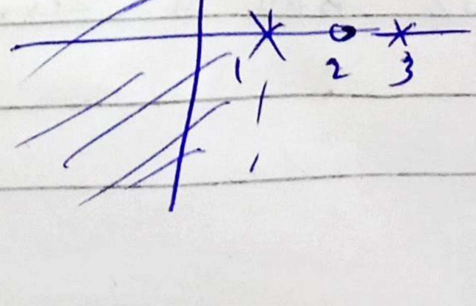
$\boxed{\Phi_{xx}(s) = X(s)\,X(-s)}$

[Time reversal yields a -ve sign in s-domain]

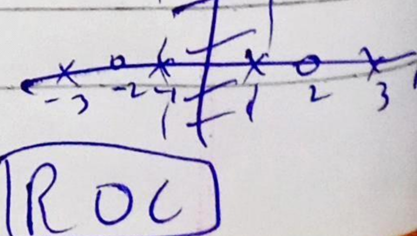
• $X(s) \Rightarrow$



$X(-s) \Rightarrow$



$\therefore$ Common Part $[X(s)X(-s)]$



$\boxed{\text{ROC}}$